

XRPL Validator — Daily Engineering Report • 2026-08-20

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **14 commits** (xrpl-validator 14)

Engineering changes

- **feat(xrpl-ledger): the flow driver — fib pace, zero-flow TER, consumed-trim stream, dead-level hydration** (bb1ff88, xrpl-validator) The whole remaining Payment class (6 both-tesSUCCESS + 3 TER, every multi-strand/pool payment on the board) closed by four receipt-pinned deltas against a skeleton that was already faithful. Design and all receipts in docs/FLOWDRIVER-DESIGN.md.
- **perf(xrpl-node): cache deterministic negative RPC lookups; share the HTTP client** (e302884, xrpl-validator) A fully warm single-fixture probe measured 77s wall / 8s CPU — 90% network wait — because the cache stored only SUCCESSFUL lookups, so every negative answer (absent line, absent object: most of what the loaders probe) was a live round-trip to the reference node on every run, and every call built its own request client (no connection reuse).
- **fix(xrpl-ledger): NFT accept — seller-owns-token NoPermission, IOU-priced settlement, strict page bound** (527b389, xrpl-validator) The whole NFTTokenAcceptOffer cluster, three receipt-pinned rules: 1. "The seller must own the token" (NFTTokenAcceptOffer.cpp:229; buy-arm :170): a stale offer whose owner no longer holds the NFToken is tecNO_PERMISSION — our transfer_token surfaced it later as tecNO_ENTRY. #106333939 2FEB03EC accepts a sell offer whose seller parted with 00081B58...0100. Checked in both direct and brokered arms.
- **fix(xrpl-ledger): AMMWithdraw — LP line survives at zero unless default; issuer dir-tail hydration** (6f72c88, xrpl-validator)
- A plain withdraw burns LP tokens via redeemIOU — the ORDINARY line machinery — so trustDelete's default-state conditions apply: the LP-side limit must be zero and no quality settings present, else the zeroed LPToken line SURVIVES as Modified. (deleteAMMTrustLine, which deletes unconditionally, runs only when the AMM itself is deleted — Transactor.cpp:1050.) Our tear_down_lp_line's own comment stated the ...
- **fix(xrpl-ledger): engine-placed offers carry BookDirectory/BookNode/OwnerNode hints** (2b1fc05, xrpl-validator) D63363BB (l106030993) + FAFBE1529 (l106065267): a bot places an offer and cancels it six txs later in the same ledger. Our placement wrote the Offer object WITHOUT its directory hints; OfferCancel reads BookDirectory off the object to unlink the book page, found None, and silently skipped the removal — the emptied page (CE26A7EB...:2) survived. Offers hydrated from mainnet carry the hints, which is why only ...
- **fix(xrpl-ledger): four singles — CredentialAccept, SignerListSet, TrustSet, CheckCash** (7965e2d, xrpl-validator) CredentialAccept (CredentialAccept.cpp:122-123): expired credential is deleted (both dir unlinks, issuer OwnerCount-1) instead of accepted; on accept the reserve moves issuer->subject when they differ.

- **feat(xrpl-ledger): full PaymentChannel port — tfClose/tfRenew, expiry, keylet, hints** (56832be , xrpl-validator) A4E67EFD (1106124864): a pure tfClose claim (Flags only, no Balance) was temMALFORMED because do_apply demanded a Balance. Port the complete PaymentChannelClaim.cpp: expiry-close FIRST (any claimant closes an expired channel, before the permission gate), Balance arm (dst needs a channel authorization — tecNO_PERMISSION under fixCleanup3_2_0 — sig key must match the channel's, tecUNFUNDED_PAYMENT both directions, ...
- **feat(xrpl-ledger): MPT subsystem — direct MPT Payments and MPT Clawback** (121869c , xrpl-validator) 4F58CD19 (1106134471) and 3EC225FD (1106259185) both died in amount parsing: nothing recognized {mpt_issuance_id, value}. New tx::mpt module ports the MPTokensV1 subset: - keylets: mptIssuance = sha512h(0x007E || MPTID24), mptoken = sha512h(0x0074 || issuance_KEY || holder) — keyed by the issuance's key, not its id. - Payment direct arm (Payment.cpp:516-590; V1 MPT never routes through flow): requireAuth both ...
- **feat(xrpl-ledger): anchored-slice bridged legs — single-path pool admission at SPOT, execution capped at the tip** (4ee4288 , xrpl-validator) #106093637 BDB95F8A, the last parked OfferCreate. tfPassive sell of 1908.4258 RLUSD for 0.02948 BTC, autobridged. After the first pass consumes the 386EAF9 leg-A head, mainnet's next iteration admits the strand at ub 64717.04 = leg-A tip 1.06832e-6 x leg-B pool SPOT 6.05785e10 (FLOWDBG receipt) and fills 12.03 RLUSD through the rURtT5MM clip offer plus the pool. Our bridged upper bound priced leg B at its FIB rate, ...
- **fix(xrpl-ledger): AMM offer generation reads pool balances ZERO-IF-FROZEN** (29b5ccb , xrpl-validator) Restock batch 2026-08-20 (24 fresh ledgers): 4 divergent, all ONE family — an arb bot's circular XRP->BITx->REGARD payments (F3BD86FA et al. across 1106423982/1106427147/1106428943/1106428898, 24 txs) where mainnet says tecPATH_DRY and we delivered (or tecPATH_PARTIAL'd). The FFI trace names it in rippled's own words: 'AMMLiquidity::getOffer, frozen accounts' — fetchBalances reads through ammHolds(fhZERO_IF_FROZEN), ...
- **fix(xrpl-ledger): model ISSUER-ONLY path elements (type 32) as same-currency cross-issuer books** (603a322 , xrpl-validator) #106425462 CF3FFF81: circular arb XRP → USD.rvYA → USD.rhub8 → XAH.rswh whose path carries {issuer: rhub8, type: 32} between two account hops. rippled's toStrand keeps the running currency and emits a USD.rvYA → USD.rhub8 BOOK step (the trace shows it crossed: offer 576ED819 + two pools) — our path_legs required a currency on every element and dropped the whole path, answering tecPATH_DRY where mainnet delivers 16.328 XAH.
- **fix(xrpl-ledger): payment-mode AMM fills pay the IN-side transfer rate GROSS** (1e9a2e4 , xrpl-validator) #106426837 6BC6ACF9: circular XRP → FLR → XRPLOL arb through two pools, both tes but 11v8 — every fib fill 0.3% short (ours 1222.247 per slice vs mainnet's 1225.914; the arithmetic pins FLR's TransferRate 1.003 exactly). rippled's rdrIn: the previous step redeems, so the strand parts with ofrIn × rate while the pool receives the net — the trace's 9.931638943875375 = 9.901933144442048 × 1.003 to the digit. Our ...
- **feat(xrpl-ledger): last-LP withdrawal deletes the AMM — lines, object, pool account** (aa37f6b , xrpl-validator) #106430239 419A5D2C: tfWithdrawAll by the pool's ONLY LP. rippled's deleteAMMAccountIfEmpty fires once LPTokenBalance hits zero: every pool asset trust line is deleted UNCONDITIONALLY (deleteAMMTrustLine — the survive-at-zero rule is for ordinary withdrawals only), the AMM object leaves the pool's owner directory, and the pool AccountRoot is erased — the trace trustDeletes all three lines then threads to the ...
- **fix(xrpl-node): book hydration follows ledger_data markers — 400 pages, budgeted offers** (64c8246 , xrpl-validator) #106433073 462DE605: a tfImmediateOrCancel selling 4845 drops for RLUSD. The maker's page sat

50 QUALITY LEVELS past the parent tip of the bot-deep RLUSD/XRP book; the trailing enumeration was one 24-row ledger_data call, so the walk saw clob=None and killed an IoC mainnet fills (tecKILLED vs tesSUCCESS, the maker offer well inside our own threshold).

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Stealth Payment Destinations** (privacy) Add an optional sfStealthScanKey SField to AccountRoot (published via a new AccountSet flag) and a new sfEphemeralPubKey SField on Payment, following
- **Amendment Implementation Bounty Vault** (governance) Add a new ltAMENDMENT_BOUNTY object (owner-reserved, keylet::amendmentBounty(amendmentID)), funded by any account via AmendmentBountyFund into a Singl
- **AMM Concentrated Liquidity Price Bands** (amm-defi) Add a new ltAMM_LIQUIDITY_POSITION object (owner-reserved, keylet::ammPosition(ammID, account, sequence)) and an AMMDepositRange transaction carrying

Appendix — commits

Time	Repo	Commit	Subject
01:40	xrpl-validator	bb1ff88	feat(xrpl-ledger): the flow driver — fib pace, zero-flow TER, consumed-trim stream, dead-level hydration
11:26	xrpl-validator	e302884	perf(xrpl-node): cache deterministic negative RPC lookups; share the HTTP client
11:45	xrpl-validator	527b389	fix(xrpl-ledger): NFT accept — seller-owns-token NoPermission, IOU-priced settlement, strict page bound
11:55	xrpl-validator	6f72c88	fix(xrpl-ledger): AMMWithdraw — LP line survives at zero unless default; issuer dir-tail hydration
12:56	xrpl-validator	2b1fc05	fix(xrpl-ledger): engine-placed offers carry BookDirectory/BookNode/OwnerNode hints
12:56	xrpl-validator	7965e2d	fix(xrpl-ledger): four singles — CredentialAccept, SignerListSet, TrustSet, CheckCash
15:08	xrpl-validator	56832be	feat(xrpl-ledger): full PaymentChannel port — tfClose/tfRenew, expiry, keylet, hints
15:20	xrpl-validator	121869c	feat(xrpl-ledger): MPT subsystem — direct MPT Payments and MPT Clawback
15:45	xrpl-validator	4ee4288	feat(xrpl-ledger): anchored-slice bridged legs — single-path pool admission at SPOT, execution capped at the tip
16:41	xrpl-validator	29b5ccb	fix(xrpl-ledger): AMM offer generation reads pool balances ZERO-IF-FROZEN
17:26	xrpl-validator	603a322	fix(xrpl-ledger): model ISSUER-ONLY path elements (type 32) as same-currency cross-issuer books
17:38	xrpl-validator	1e9a2e4	fix(xrpl-ledger): payment-mode AMM fills pay the IN-side transfer rate GROSS

Time	Repo	Commit	Subject
17:43	xrpl-validator	aa37f6b	feat(xrpl-ledger): last-LP withdrawal deletes the AMM — lines, object, pool account
23:20	xrpl-validator	64c8246	fix(xrpl-node): book hydration follows ledger_data markers — 400 pages, budgeted offers